

ATPL EASA STUDY GUIDE | 033 - FLIGHT PLANNING

Chapter 2: EASA IFR Fuel Policy (AIR-OPS CAT.OP.MPA.180)

Syllabus Source: CAE Oxford (pp. 41-82) | Pass Target: >= 90% AviationExam

1. THE EASA REGULATORY FUEL BREAKDOWN

Under EASA AIR-OPS (CAT.OP.MPA.180), an IFR commercial flight is legally prohibited from departing without carrying minimum certified fuel reserves:

Fuel Component	Regulatory Definition & Calculation Basis	Operational Role & Minimum Rules
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1. Taxi Fuel	Fuel consumed before take-off (APU burn, engine start, taxiing to runway).	Standard allowance: typically 150 to 300 kg for twin-jets (12-15 min).
2. Trip Fuel	Fuel required from brake release at departure to touchdown at destination.	Includes take-off, climb, cruise, descent, approach, and landing at destination.
3. Contingency Fuel	Buffer for unforeseen deviations (weather, ATC routing, lower flight level).	HIGHER OF: 5% of Trip Fuel, OR 3% with En-Route Alternate (ERA), OR 20 min flight time.
4. Destination Alternate Fuel	Fuel required to divert from missed approach at destination to alternate.	Missed approach + climb + cruise to alternate + descent + approach + landing.
5. Final Reserve Fuel	Fuel to hold at 1,500 ft AGL over alternate in ISA conditions.	MANDATORY: 30 minutes for turbine / jet aircraft (45 minutes for piston aircraft)!
6. Additional Fuel	Required only if aircraft must comply with depressurization or OEI drift down.	Ensures reaching a safe airport if engine fails or cabin depressurizes at critical point.
7. Extra Fuel	Discretionary fuel loaded at Commander's authority for anticipated delays.	Added for holding, thunderstorm circumnavigation, or tankering.

2. THE MASTER FUEL EQUATIONS

- Minimum Required Take-Off Fuel: $\text{Take-Off Fuel} = \text{Trip Fuel} + \text{Contingency Fuel} + \text{Alternate Fuel} + \text{Final Reserve Fuel} (+ \text{Additional Fuel})$.
- Minimum Required Block Fuel (Ramp Fuel): $\text{Block Fuel} = \text{Taxi Fuel} + \text{Minimum Required Take-Off Fuel} (+ \text{Extra Fuel})$.

STEP-BY-STEP PROBLEM SOLVING (FOR DUMMIES / PASO A PASO)

[+] STEP-BY-STEP ALGORITHM / PASO A PASO - Step-by-Step Algorithm (Dummies Guide): IFR Fuel Planning & Minimum Le

Problem: A320 flight. Trip Fuel = 6,000 kg. Holding fuel flow at 1,500 ft = 40 kg/min (2,400 kg/hr). Alternate Fuel = 1,400 kg. Taxi = 200 kg. Step 1: Calculate Contingency Fuel (GREATER of 5% Trip or 5 min holding): - 5% Trip = $0.05 \times 6,000 = 300$ kg. - 5 min holding = $5 \times 40 = 200$ kg. -> Contingency = 300 kg. Step 2: Calculate Final Reserve (30 min for jet airliners): - Final Reserve = $30 \times 40 = 1,200$ kg (untouchable!). Step 3: Take-Off Fuel TOF = Trip (6,000) + Contingency (300) + Alternate (1,400) + Final Reserve (1,200) = 8,900 kg. Step 4: Minimum Ramp Fuel = TOF + Taxi = $8,900 + 200 = 9,100$ kg.

3. STEP-BY-STEP WORKED EXAMPLE (FOR DUMMIES / PASO A PASO)**[*] OFFICIAL DEFINITION - PRACTICAL CALCULATION: Compiling an EASA IFR Fuel Plan**

SCENARIO (Classic ATPL Flight Planning Exam Question): An Airbus A320 is planned for an IFR passenger flight from Madrid to Paris: - Planned Taxi Fuel: 200 kg - Calculated Trip Fuel: 5,000 kg - Diversion fuel to Paris Alternate (Orly to Beauvais): 1,200 kg - Holding fuel consumption at 1,500 ft AGL: 2,400 kg/hour - No En-Route Alternate (ERA) is designated (standard 5% contingency applies) - Commander requests 300 kg Extra Fuel for holding QUESTION: Calculate (1) Contingency Fuel, (2) Final Reserve Fuel, (3) Minimum Take-Off Fuel, and (4) Total Block Fuel. SOLUTION PASO A PASO (FOR DUMMIES): Step 1: Calculate Contingency Fuel (Standard 5% of Trip Fuel): - Contingency = 5% of 5,000 kg = $0.05 \times 5,000$ kg = 250 kg. Step 2: Calculate Final Reserve Fuel (30 minutes of holding for jet!): - Holding fuel burn = 2,400 kg/hour. - Final Reserve (30 min = 0.5 hour) = $0.5 \text{ hr} \times 2,400 \text{ kg/hr} = 1,200$ kg! Step 3: Calculate Minimum Required Take-Off Fuel: - Take-Off Fuel = Trip + Contingency + Alternate + Final Reserve - Take-Off Fuel = $5,000 \text{ kg} + 250 \text{ kg} + 1,200 \text{ kg} + 1,200 \text{ kg} = 7,650$ kg. Step 4: Calculate Total Required Block Fuel (Add Taxi and Extra Fuel): - Block Fuel = Taxi (200 kg) + Take-Off Fuel (7,650 kg) + Extra (300 kg) - Block Fuel = $200 + 7,650 + 300 = 8,150$ kg! FINAL ANSWER: Contingency = 250 kg; Final Reserve = 1,200 kg; Min Take-Off Fuel = 7,650 kg; Total Block Fuel = 8,150 kg.

[!] EXAM TRAP (AVIATIONEXAM) - Final Reserve: 30 Min Turbine vs 45 Min Piston Trap

- For TURBINE / JET aircraft: Final Reserve is 30 MINUTES holding at 1,500 ft. - For PISTON aircraft: Final Reserve is 45 MINUTES holding! - Final Reserve is SACROSANCT: If you calculate that you will land with less than Final Reserve, you MUST DECLARE 'MAYDAY MAYDAY MAYDAY FUEL'!

4. AVIATIONEXAM PRACTICE QUESTIONS (WITH EXPLANATIONS)

Q1: Under EASA AIR-OPS, what is the minimum duration of FINAL RESERVE FUEL required for a commercial t

- [A]: 45 minutes
- [B]: 30 MINUTES holding at 1,500 ft AGL above the alternate aerodrome under ISA conditions
- [C]: 20 minutes
- [D]: 60 minutes

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. For turbine-powered aeroplanes, final reserve fuel is calculated for 30 minutes of holding at 1,500 ft (450 m) above aerodrome elevation in standard ISA conditions.

Q2: What is the standard CONTINGENCY FUEL percentage required on an IFR flight when no en-route alternate

- [A]: 10% of Trip Fuel
- [B]: 5% OF TRIP FUEL (or 5 minutes, whichever is greater)
- [C]: 3% of Trip Fuel
- [D]: 20%

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. The baseline contingency fuel required by EASA CAT.OP.MPA.180 is 5% of the planned trip fuel (reduced to 3% only if an en-route alternate is nominated).

Q3: When MUST a flight crew legally declare 'MAYDAY MAYDAY MAYDAY FUEL' to Air Traffic Control?

- [A]: When diverting to any alternate
- [B]: When the estimated usable fuel upon landing at the nearest suitable aerodrome is calculated to be LESS THAN the planned Final Reserve Fuel
- [C]: When contingency fuel is exhausted
- [D]: On any missed approach

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. Under ICAO Annex 6 and EASA regulations, MAYDAY FUEL is an explicit declaration of emergency that must be transmitted whenever calculated fuel on touchdown is less than the mandatory 30-minute final reserve.